Factor-base designs: A11 replication on p_2 = 2013265921 (run-20261015T2307Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md §5, with Amendment 2 F1 (registered on `research/factor-base-designs-lean` in commit `e0afa082d`, before any p_2 row) and Amendment 1 for everything F1 does not change. This is F1 only. F2–F4 were not run here.

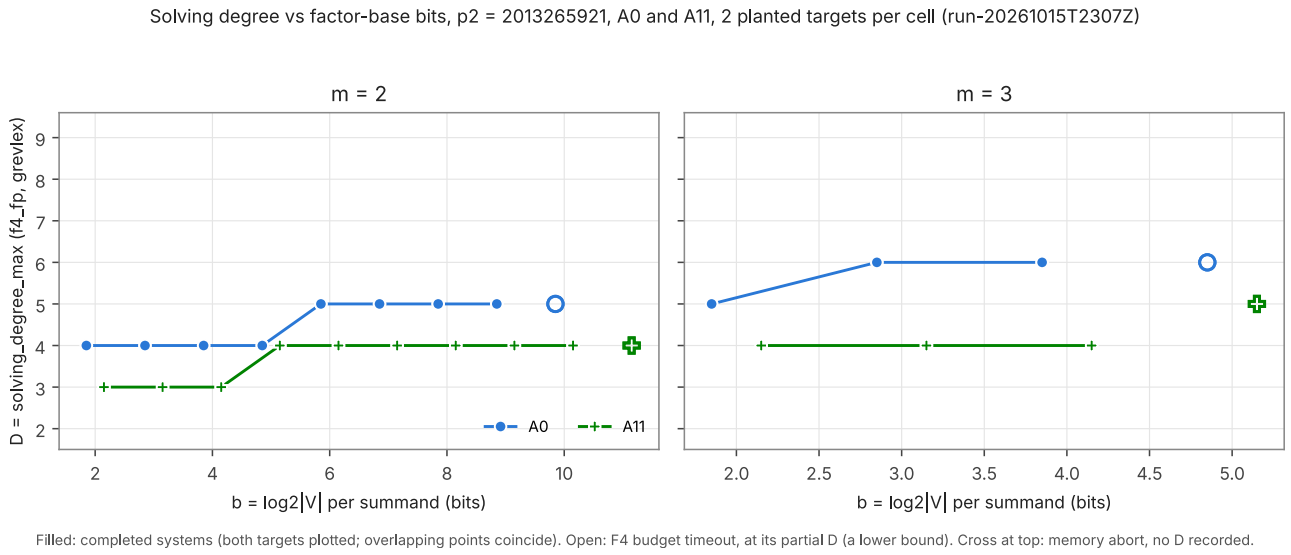
Run

- **Command:** `target/release/examples/fb_designs_slope run --p 2013265921 --arms A0,A11 --m 2,3 --out-dir research/factor_base_designs_20261015/runs/p2` (defaults: t from 2, at most 27 = $v_2(p_2 - 1)$, 2 planted targets per cell, F4 budget 300 s, hard kill 420 s, `ulimit -v 13000000` per child).
- **Commit measured:** `d5f675260867f2d025e080e65d8d7027431bc845`, recorded in every row.
- **Host:** "vm", Intel Xeon @ 2.10 GHz, 4 cores, a Claude Code cloud container. This is a different container and CPU from the p_1 run (2.80 GHz), so times are not comparable across the two runs. **Isolation:** one system at a time, no other work running.
- **Window:** 2026-10-15 23:07Z to 2026-10-16 about 00:08Z (UTC).
- **Seeds (Amendment 2):** `0x46424453 ⊕ prime≪48 ⊕ arm≪40 ⊕ m≪32 ⊕ t≪16` with p_2 = 1. Target k is planted from `cell ⊕ (k+1)≪8`. With p_1 = 0, the p_1 seeds are unchanged: A0 and A11 at m = 3, t = 3 were rebuilt and match their committed rows (x_R, D, max_cols).
- **Records:** `runs/p2/{A0,A11}-m{2,3}.jsonl` and `.log`, `runs/p2/analysis.txt` (native `analyze`) and `runs/p2/instrument-check-p2.txt`.

F1 pre-check

$p_2 < 2^{31}$, and `f4_fp` reduces modulo $p < 2^{32}$. Eight planted systems at t = 2 were built before any p_2 row: A0 and A11, m = 2 and 3, both targets. Each passed the set, presentation and vanishing checks, and each kept its planted solution in F4's basis (`planted_ok` true, not inconsistent). The check is recorded in `instrument-check-p2.txt`.

Graphs



Filled: completed systems (both targets plotted; overlapping points coincide). Open: F4 budget timeout, at its partial D (a lower bound). Cross at top: memory abort, no D recorded.

Figure 1: Measured D against b on p_2, A0 and A11, m = 2 (left) and m = 3 (right). Both targets of each cell are plotted (n = 2 per cell; they coincide). Open markers are systems stopped by the 300 s F4 budget, at their partial D. Rows from `runs/p2/*.jsonl`.

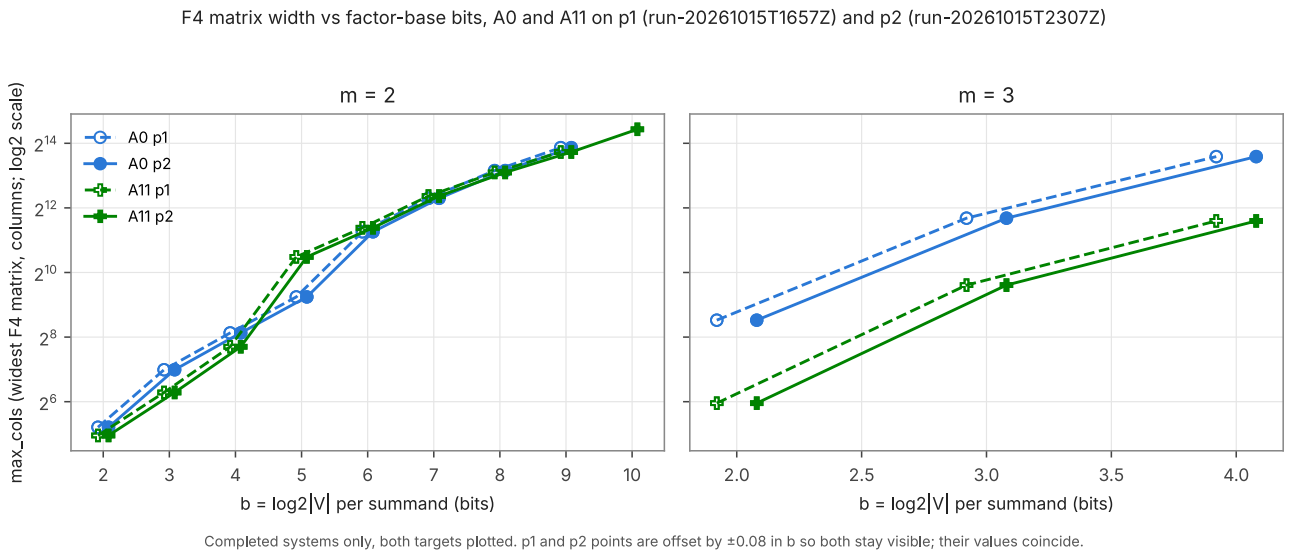


Figure 2: Widest F4 matrix (`max_cols`, $\log_2$ scale) against b, A0 and A11 on p_1 (dashed, hollow) and p_2 (solid). The points are offset ±0.08 in b so both primes stay visible. Completed systems only.

F1 decision rule (registered for m = 3 on p_2)

"A11 replicates if A11's cell D ≤ A0's cell D − 1 and A11's `max_cols` < A0's at every shared completed t, with at least 3 shared completed cells." The cell `max_cols` here is the mean over the cell's two systems.

t	A11 D	A0 D	A11 max_cols	A0 max_cols	condition
2	4	5	62	368	met
3	4	6	781	3285	met
4	4	6	3096	12314	met

Reading: replicates, on 3 shared completed cells, the minimum the rule accepts. At t = 5 both arms' systems hit the F4 budget, so that cell is missing for both, with partial D of 5 (A11) and 6 (A0) when stopped.

σ on p_2 (from `analysis.txt`): A0 m = 2 is 0.1905 (8 cells), A0 m = 3 is 0.5000 (3), A11 m = 2 is 0.1500 (9), and A11 m = 3 is 0.0000 (3). The registered width-growth slope $\Delta\log_2$ `max_cols`/Δb (descriptive) is 1.261, 2.532, 1.215 and 2.821, in the same order.

Observations

- **p_1 and p_2 agree system for system.** 44 planted systems completed on both primes at the same (arm, m, t, target): A0 and A11, m = 2 and 3. All 44 have the same D, and all 44 have the same `max_cols`. The two primes share no curve, set or target. At these sizes, D and matrix width do not vary with the prime or the instance.
- **m = 2 (descriptive; the rule is not registered here).** A11's D is one below A0's at t = 2–4 and 6–9, and equal at t = 5. A11 is narrower at t = 2–4 and 8–9, and wider at t = 5–7. The widest of those is t = 5: 1431 vs 606 columns.
- **Further cells.** A11 at m = 2 completed one more cell on p_2 (t = 10, D = 4, about 257 s per system) than on p_1, where t = 10 timed out at 300 s on a different CPU. A0 at m = 2 stops at t = 10 on both primes.
- **The two targets of a cell are not always distinct systems.** At small t only 2–4 elements of V lie on the curve, so both targets can draw the same abscissae. Then x_R coincides, and for the tower arms the system is identical. On p_2 this happened in A11 at t = 2 (m = 2 and 3). On p_1, eight cells at t ≤ 4 across arms have it. D is unaffected, but those cells hold one distinct system, not two. The draw rule is the registered one and was not changed.

Every system (p_2)

Target 0 / target 1. Time is F4 wall seconds. Status is ok or timeout (F4 budget of 300 s). There were no memory aborts and no planted-solution violations.

arm	m	t	vars	eqs	D	max_cols	time (s)	status
A0	2	2	4	5	4 / 4	37 / 37	0.0 / 0.0	ok, ok
A0	2	3	6	7	4 / 4	127 / 127	0.0 / 0.0	ok, ok
A0	2	4	8	9	4 / 4	280 / 280	0.0 / 0.0	ok, ok
A0	2	5	10	11	4 / 4	606 / 606	0.0 / 0.0	ok, ok
A0	2	6	12	13	5 / 5	2451 / 2451	0.2 / 0.2	ok, ok
A0	2	7	14	15	5 / 5	5031 / 5031	1.1 / 1.2	ok, ok

A0	2	8	16	17	5 / 5	9119 / 9119	9.3 / 9.0	ok, ok
A0	2	9	18	19	5 / 5	14922 / 14922	59.6 / 59.4	ok, ok
A0	2	10	20	21	5 / 5	61361 / 61361	301.1 / 301.5	timeout, timeout
A0	3	2	7	8	5 / 5	368 / 368	0.0 / 0.0	ok, ok
A0	3	3	10	11	6 / 6	3285 / 3285	0.3 / 0.3	ok, ok
A0	3	4	13	14	6 / 6	12314 / 12314	9.5 / 9.7	ok, ok
A0	3	5	16	17	6 / 6	97456 / 97456	301.0 / 301.3	timeout, timeout
A11	2	2	9	10	3 / 3	31 / 31	0.0 / 0.0	ok, ok
A11	2	3	12	13	3 / 3	78 / 78	0.0 / 0.0	ok, ok
A11	2	4	15	16	3 / 3	208 / 208	0.0 / 0.0	ok, ok
A11	2	5	18	19	4 / 4	1431 / 1431	0.0 / 0.1	ok, ok
A11	2	6	21	22	4 / 4	2673 / 2673	0.5 / 0.6	ok, ok
A11	2	7	24	25	4 / 4	5282 / 5282	2.4 / 2.3	ok, ok
A11	2	8	27	28	4 / 4	8763 / 8763	10.6 / 10.4	ok, ok
A11	2	9	30	31	4 / 4	13579 / 13579	53.9 / 53.7	ok, ok
A11	2	10	33	34	4 / 4	22118 / 22118	259.5 / 255.7	ok, ok
A11	2	11	36	37	4 / 4	184920 / 184920	302.0 / 301.9	timeout, timeout
A11	3	2	15	16	4 / 4	62 / 62	0.0 / 0.0	ok, ok
A11	3	3	20	21	4 / 4	781 / 781	0.0 / 0.1	ok, ok
A11	3	4	25	26	4 / 4	3096 / 3096	1.3 / 1.3	ok, ok
A11	3	5	30	31	5 / 5	40486 / 40486	300.8 / 301.6	timeout, timeout

Requested and completed

4 series, 27 cells, 54 planted systems: 46 completed and 8 stopped at the F4 budget (A0 m = 2 t = 10, A0 m = 3 t = 5, A11 m = 2 t = 11, A11 m = 3 t = 5, both targets each). The commit that landed these rows (`c195dfa64`) states “50 systems, 42 completed” in its message. That was a miscount; these figures are correct.

Deviations

None from Amendment 2 F1. Not run: F2 (the N11 null), F3 (the m = 3, t = 5 cell at 3600 s) and F4 (m = 4). Without F2, whether A11′s lower D comes from its structure or only from its equation shape is still open on both primes.